SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1516

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Duration: 60 Minutes** **Total Marks: 25**

Question Count: 5

Code of Conduct (192511481)

I VINO ALDRIN E (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: E. Vinoaldrin

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty **Total Marks Awarded:**

Assessment Tasks

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services..
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

ASSESSMENT TOOL-I
CO1

NAME :- VINO ALDRIN E
Roll. NO. :- 192511481
Sub. Code :- CSA1516

Cloud Computing and Big Data Analytics - CSA1516.

Q.1. Describe the evolution from Standalone Computing to Cloud-based Systems. Summarize key technological transitions that enabled Cloud Computing.

ANS :-

* Standalone computing refers to a single Computer that stores data and runs applications locally.

* As Organizations grew, they shifted to Client-Server architecture, where clients accessed resources from Centralized servers.

* Later, distributed Computing Connected multiple Computers to share Processing and Storage.

* The development of the internet enabled global connectivity, while Virtualization allowed multiple virtual machines to run on one physical server, improving resource utilization.

* These advancements led to cloud computing, where computing resources are delivered over the internet on demand.

Key technological transitions:-

* Standalone computing → Client-server computing.
* Client-Server → Distributed Computing.
* Internet & high-speed networking.
* Virtualization technology.
* Data Centers & resource pooling.

Examples :-

→ Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP) Provide cloud services accessible from anywhere.

Q.2. Interpret the working of hypervisors in Virtualization. Differentiate b/w Type 1 and Type 2 hypervisors with examples.

* A hypervisor is software that creates and manages multiple Virtual Machines on a single physical computer.

* It allocates CPU, memory, Storage, and network resources to each VM while keeping them isolated.

Difference b/w Type 1 and Type 2 Hyper-Visors.

Type 1 Hypervisor	Type 2 Hypervisor
* Runs directly on hardware.	* Runs on top of a host OS.
* Faster & more secure.	* Slightly slower due to host OS.
* Used in data centres and cloud.	* Used for personal computers & testing.
* Example:- VMware, ESXi, Microsoft Hyper-V, Xen.	* Example:- Oracle VirtualBox, VM ware Workstation.

Q.2. Describe the role of APIs in Cloud service integration. Explain how APIs support interoperability b/w services.

⟷ An Application Programming Interface (API) is a set of rules that allows different software applications to communicate.

*→ In Cloud Computing, APIs enable different cloud services and applications to exchange data and perform operations automatically.

*→ APIs Support Interoperability by allowing services from different providers to work together.

*→ They enable ~~at~~ automation, Secure communication, integration of applications, and data sharing without requiring changes to the underlying systems.

Benefits:-

* Easy integration b/w cloud services.
* Automation of tasks.
* Secure data exchange.
* Better Scalability & flexibility.

Example:- A web application can use Google Maps API for location services and Stripe API for online payments while being hosted on AWS.

Q.4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.

Ans→ *→ Resource pooling is a cloud computing feature where computing resources such as servers, storage, memory, and networks are shared among multiple users.

*→ The cloud provider dynamically allocates resources based on user demand while keeping each user's data isolated.

Example: In Microsoft Azure, several organizations share the same physical servers, but each organization uses its own virtual machines securly without affecting others.

Q.5. Interpret the concept of rapid elasticity in Cloud computing. Explain how it supports dynamic workloads.

- Rapid elasticity is the ability of cloud computing to automatically increase or decrease computing resources based on workload demand.

- Users can quickly scale resources up during high demand and scale them down when demand decreases.

This supports dynamic workloads by maintaing application performance while reducing unnecessary costs. Organizations pay only for the resources they use.

Benefits:-

* Automatic Scaling
* Cost efficiency.
* Improved Performance.
* Better user experience.
* Handles sudden traffic increases.

Example:- During an online shopping festival, Amazon automatically adds more cloud servers to handle increased customer traffic and removes them after the sale ends.